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Assignment: Part 5: Modeling Credit Card Default Risk and Customer Profitability

Submit by December 13, 11:59 PM PT

Ready to start your next assignment?

You'll find instructions and a place to submit below.

Instructions (/learn/analytics-excel/peer/sDcrr/part-5-modeling-credit-card-default-risk-and-customer-profitability)

My submission (/learn/analytics-excel/peer/sDcrr/part-5-modeling-credit-card-default-risk-and-customer-profitability/submit)

Discussions (/learn/analytics-excel/peer/sDcrr/part-5-modeling-credit-card-default-risk-and-customer-profitability/discussions)

This is a preview of your project. Changes have been saved but not submitted.

Modeling Credit Card Default Risk and Customer Profitability

December 13, 2015

What was the binary classification model you developed in Final Quiz Part 1, Question #1? Type your model in the box below.

- The **binary classification model** is $\text{Score} = -0.5 * (\text{years at employer}) - 0.5 * (\text{credit card debt})$.
- The output variable **Scores range** is $[-3.5, 3.5]$.
- The **threshold score** that **minimizes costs per event** on the **training set** is **0.2**.
- We have **binary dependent variable default** defined as follows from the **classification output variable Score**:

default = 1, if Score > threshold score

default = 0, otherwise.



- Model's **AUC** on the **training set** is **0.82**.
- Model's **AUC** on the **test set** is **0.83**.

At the threshold score for the training set, what is the cost per event on the test set? Refer back to Final Project Quiz 1, question 6 for your answer.

Note: if you find that your costs per event on the test set are much higher than your costs per event on the training set, you should consider making your model simpler – probably using fewer input variables – as it is probably still over-fitting the training set data. Problems that are were not obvious at the ROC-curve stage may emerge when minimizing costs.

- The **threshold score** that **minimizes costs per event** on the **training set** is **0.2**.
- The **cost per event** on the **training set** with the **threshold score (0.2)** is **763\$**.
- The **cost per event** on the **test set** with the **threshold score (0.2)** is **800\$**.

Your boss asks you to recommend a standard procedure for selecting future credit card applicants. Based on internal data alone (*not using the Eggertopia scores*) and the test set performance of your two models (the binary classification model from Final Project Quiz 1 and the linear regression model from Final Project Quiz 4), which method would you recommend - binary classification, linear regression or some combination or modification of the two? Explain why you recommended what you did, referencing at least two topics/points from the course.

I shall prefer to use the **multivariate linear regression model** (from Final Project Quiz 4), since

- Both the **binary classification model** and the **multivariate linear regression model** are **linear models**, but the later computes the coefficients using the best least square fit (**minimizing the error** on the **training set**), where the former does not.
- The **information gain** for the **binary classification model** is **0.1376** in average **bits per event** (from quiz 3), whereas the **information gain** for the **multivariate linear regression model** is **0.77** in average **bits per event** (from quiz 4). Hence, the multivariate regression model provides much higher reduction in uncertainty.
- The **multivariate linear regression model** directly models the **profitability** of the customers, which provides a more **actionable (prescriptive) insight**, for instance allowing the bank to compute (**90% confidence interval** for) the **profitability estimate** for a group of customers using the **multivariate linear regression model** that we can't do using the **binary classification model**. As pointed out in quiz 4, just predicting whether a customer is **defaulter** or not using the **binary classification model** may not always be very useful to the bank from **profitability** point of view, since some defaulters may / may not be profitable to the bank, same holds for the non-defaulters as well.

Edit (peer/sDcrr/part-5-modeling-credit-card-default-risk-and-customer-profitability/submit)

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[for final project quiz 4\)](https://accounts.coursera.org/interpersonal/help?return_to=https://learner.coursera.help/hc) (https://accounts.coursera.org/interpersonal/help?return_to=https://learner.coursera.help/hc)

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